

Inventory Optimization & Supply Chain Strategy

Project Documentation

This project applies supply chain analytics and inventory management concepts to analyze demand patterns, identify inventory risks, and recommend practical reorder and safety stock strategies.

Project Objective

The objective is to transform transaction data into management insights that support better inventory decisions, reduce stockout risks, and improve inventory coverage.

Business Approach

The analysis focuses on demand behavior, product performance, demand variability, product segmentation, safety stock planning, reorder point design, and management actions.

Key Management Insights

- Identified high-demand products requiring priority attention.
- Evaluated demand variability to understand inventory uncertainty.
- Developed inventory recommendations using safety stock and reorder point concepts.
- Created an action-oriented prioritization framework for decision makers.

Recommendations Framework

Products are categorized according to demand volume and variability. Management can focus resources on products with high demand and high uncertainty while maintaining efficient inventory coverage for other categories.

Analysis Visualizations

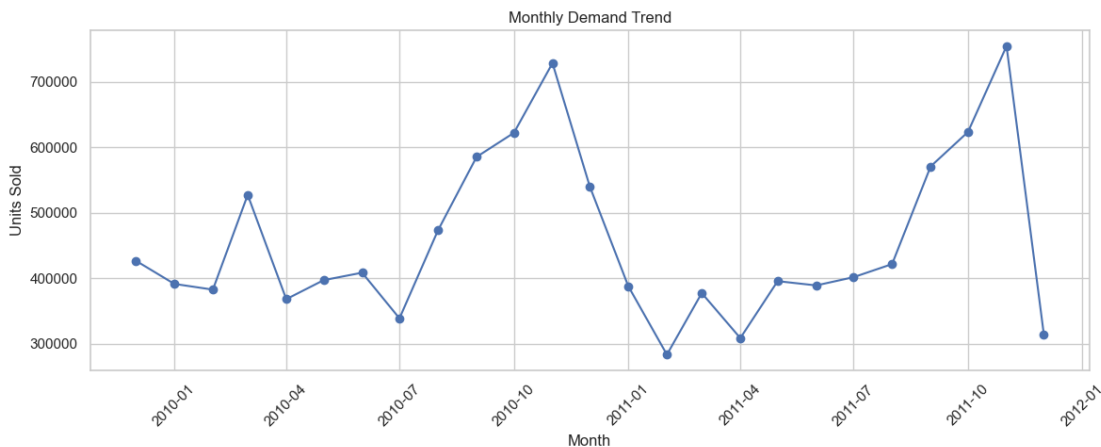


Chart 1

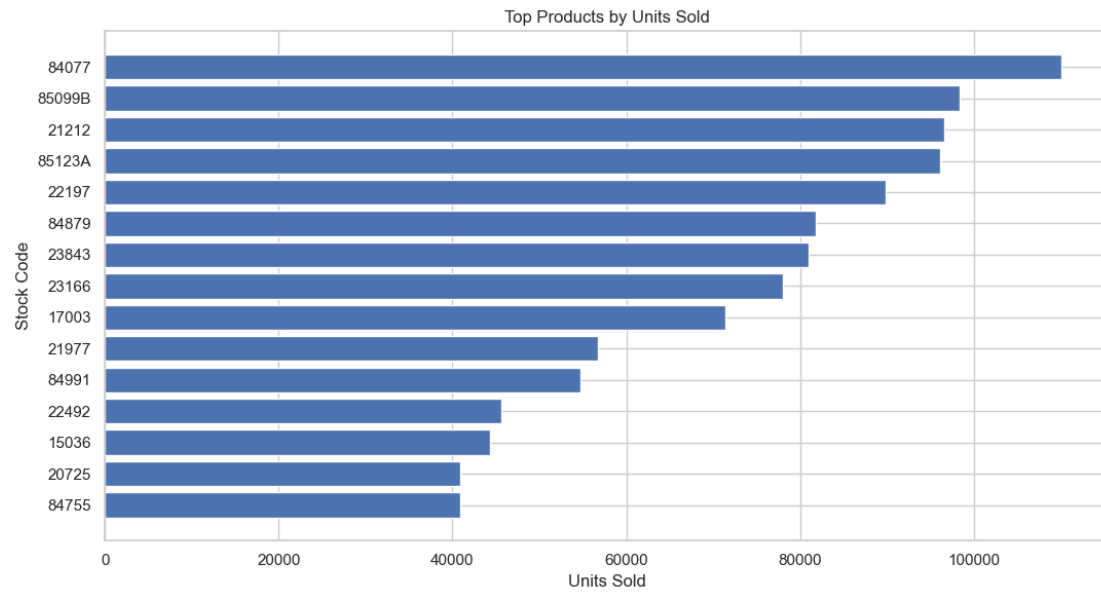


Chart 2

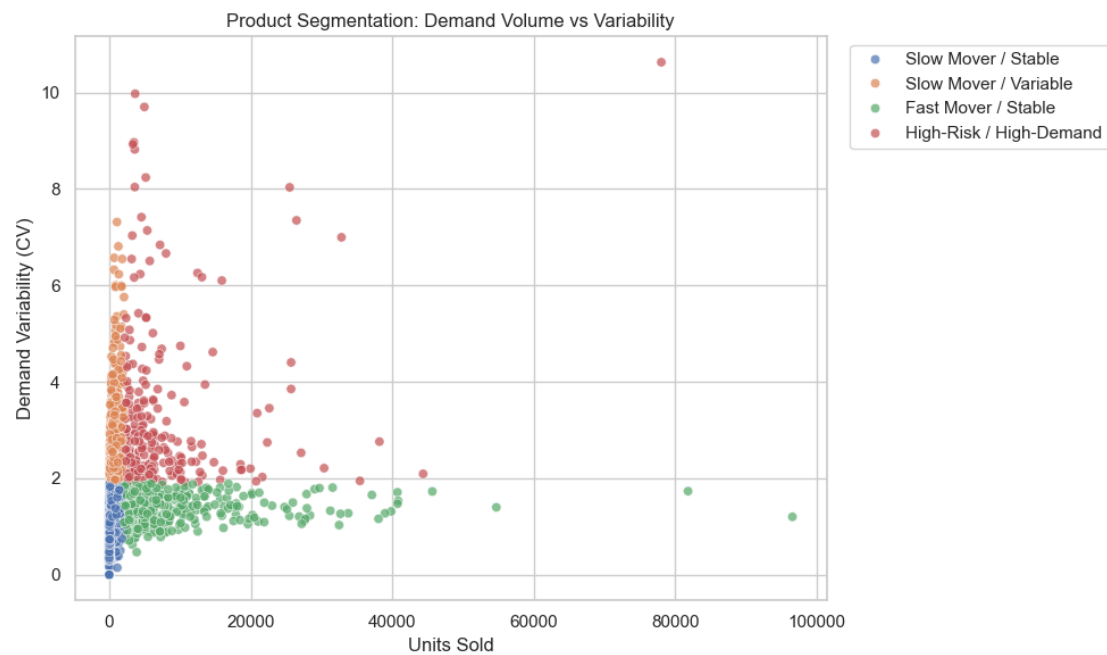


Chart 3

Business Takeaways

The project demonstrates how analytics can support operational decision making. The final output is designed for managers who need clear recommendations rather than technical implementation details.